

- 1 A child kicks a ball so that it leaves horizontal ground with a velocity of 28 m s^{-1} at an angle of 34° to the horizontal, as shown in Fig. 1.1.

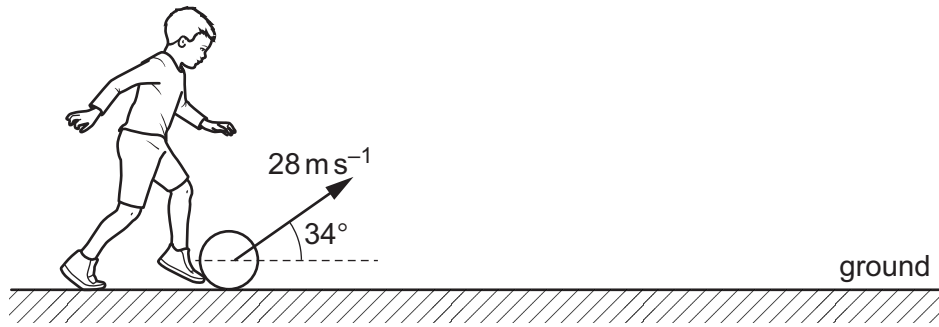


Fig. 1.1

Air resistance is negligible. The ball leaves the ground at time $t = 0$.

- (a) (i) Calculate the horizontal component v_H and the vertical component v_V of the velocity of the ball immediately after it has left the ground.

$$v_H = \dots\dots\dots \text{ m s}^{-1}$$

$$v_V = \dots\dots\dots \text{ m s}^{-1}$$

[2]

- (ii) Show that the ball reaches its maximum height at time $t = 1.6 \text{ s}$.

[1]

